that the smallest explosion which could be detected by broad-band seismographs in the UK would be about 50 kt, which is about as far as earthquake simulation technique can be taken, depending as it does on several narrow-band instruments detecting the smaller (by a factor of 10) explosions.

We propose that the broad-band seismograph looks promising as a safeguard against attempts to camouflage moderately large underground tests with a carefully timed series of explosions. It seems likely that the detection level at most stations could be better than 50 kt after array processing (experiments are in hand), though according to Kondorskaya and Aranovich[4] the resolution of the period discriminant is likely to be weaker at lower magnitudes. Nonetheless a serious would-be violator, in allowing for the possible occurrence of exceptional recording conditions, unknown improvements in techniques and the possible existence of undisclosed stations, would have to credit

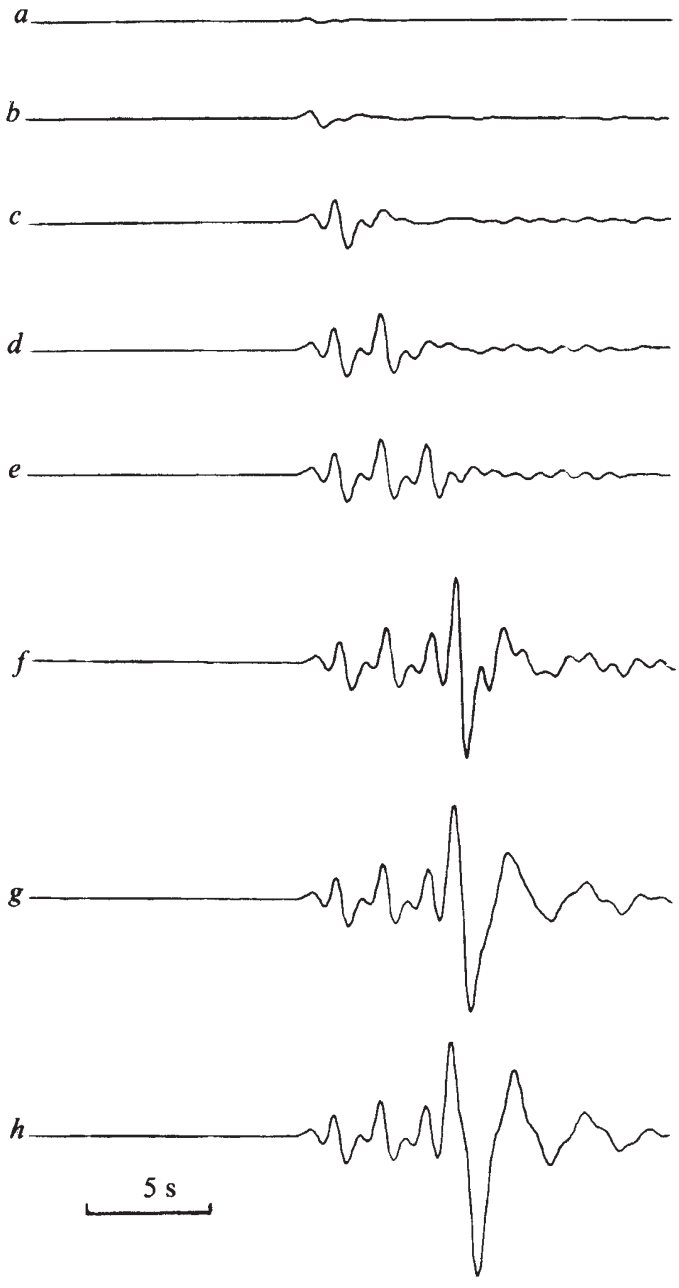


Fig. 4 Model simulation of the multiple explosion, illustrating the development of the final signal. *a*, 3 kt; *b*, + 9 kt, delayed 0.3 s; *c*, + 29 kt, delayed 1.1 s; *d*, + 29 kt, delayed 2.9 s; *e*, + 29 kt, delayed 2.7 s; *f*, + 100 kt, delayed 5.7 s; *g*, + 100 kt, delayed 6.0 s; *h*, + 100 kt, delayed 6.3 s.

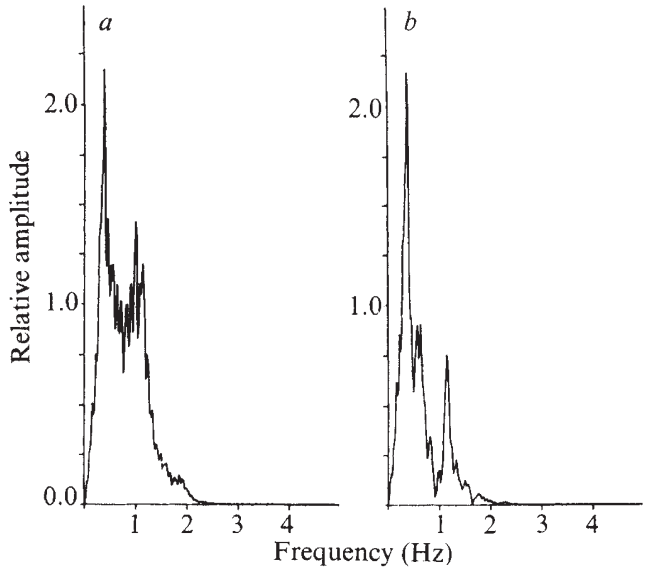


Fig. 5 *a*, Amplitude spectrum of the single explosion used in the simulation experiment; *b*, amplitude spectrum of the resultant multiple explosion. Note that the large, low frequency, spectral peak remains at the same frequency.

'national networks' with somewhat better detection capacities than publicly revealed. A safety factor of two would not be an unreasonable margin to allow. Thus, the geophysical uncertainties operate against the violator.

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Anomalous geomagnetic field during the late Ordovician

MEASUREMENTS of the Lower Palaeozoic magnetic field in the British Isles[1] show that it changed from around $D=355°$, $I=-50°$ to $D=20°$, $I=-54°$ (corresponding to an apparent shift of the south palaeomagnetic pole from 0°W to 35°W along the present Equator) between Ordovician and Devonian times. Following unpublished work by W. E. Tremlett and J. C. Briden, we report here conflicting evidence from the intrusive rocks between Conway and Pwllheli, North Wales. These rocks therefore seem to show either previously unrecognised tectonic rotations or hitherto unsuspected variation in the palaeomagnetic field.

Throughout North Wales, an area of complex Palaeozoic tectonism, a large number of acid igneous intrusions occur whose ages within the Lower Palaeozoic are ill defined. Some bear strong petrogenetic affinities with the Ordovician acid extrusive rocks of the area and on field evidence there is little doubt that such intrusions as the Tan-y-Grisiau microgranite, the Myndd Mawr microgranite, the Penmaenbach rhyolitic intrusion and the Cader Idris granophyre[2-5] are of Ordovician age, and it has commonly been inferred that the bulk of acid igneous activity in the region is coeval. On the other hand, the view is held by some workers that

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Table 1 Fisher statistics of a.f. cleaned remanences of palaeomagnetic sites from Lower Palaeozoic igneous rocks of North Wales

Rock unit	Approximate grid reference	Peak a.f. (mT)	N (sites)	R	k	α_{95}	D*	I*	Palaeomagnetic pole Long.	Lat.	$d\psi$	$d\chi$
Penmaenmawr granodiorite	SH 700 760	10–40	12	11.6	27	8	164	−71	104E	81S	13	15
Penmaenbach rhyolitic intrusion	SH 750 780	20	12	11.5	21	10	125	−71	112E	59S	15	17
Foel Fras granodiorite	SH 699 695	20	4†	3.9	214	5	207	−78	209E	71S	9	9
Garnfor granodiorite	SH 360 460	20–50	9	8.7	30	13	136	−62	84E	60S	11	14
Carreg-y-Llam granodiorite	SH 340 440	40–80	5	4.9	37	13	126	−71	111E	59S	19	22
Bodeilias granodiorite	SH 320 420	20–60	6	5.8	27	19	120	−55	87E	45S	13	19
Myndd Nefyn granodiorite	SH 320 410	30–80	7	6.5	11	14	139	−36	55E	44S	12	20
Gurn Ddu granodiorite	SH 400 470	20–60	6	5.8	22	9	244	−72	233E	55S	22	25
Yr Eifl microgranite	SH 350 450	20	6†	5.9	62	14	82	−65	122E	32S	15	19
Cader Idris basalts	SH 710 146	20–60	10	9.4	13	14	208	−73	229E	73S	18	22
Overall mean of above results (all sites)			69	63.1	11	5	150	−69	96E	72S	8	9
Overall mean of above results (excluding Myndd Nefyn and Cader Idris)			52	50.0	17	5	143	−71	108E	68S	8	9

*All mean directions quoted are *in situ* except for that of the Cader Idris basalts which has been tilt-adjusted by simple rotation about the present strike.

†Statistics presented are the within-site result from a single site at Foel Fras and Yr Eifl.

The notation used is as follows: R is the resultant of N unit vectors, k ($\simeq N-1/N-R$) is the estimate of Fisher's[13] precision κ, α_{95} is the semi-angle of the cone of 95% confidence about the mean direction D (declination in degrees east of north) and I (inclination from horizontal, positive indicating downwards), $d\psi$ and $d\chi$ are the semi-angles subtended by the major and minor axes of the ellipse of 95% confidence about the mean pole position.

some intrusions belong to a late stage of the Caledonian orogeny. In particular, W. E. Tremlett has argued that the granodiorite complex of northern Lleyn is associated with NE–SW Caledonoid faults of that area, and is therefore probably of early Devonian age; it is geochemically distinct from the Ordovician acid rocks[7–9]. He has also argued on field evidence that the Penmaenmawr microgranodiorite is post-orogenic, and in disagreement with Davies[6] assigns a similar age to the Foel Fras granodiorite 6 km south of Penmaenmawr (W. E. Tremlett, personal communication). The Penmaenmawr intrusion has given a K–Ar age of 375 Ma[10] from an admittedly weathered sample. The Lleyn granodioritic intrusions have yielded Hercynian ages (R. M. McIntyre, personal communication) but there is no geological evidence to suggest massive igneous activity in North Wales at that time, and these apparent radiometric ages are therefore to be regarded as minimum estimates.

Many of the intrusions have been sampled during our new study and the palaeomagnetism of those possessing stable remanences after a.f. cleaning is reported here (Table 1, Fig. 1). Typically at least six independently orientated specimens were obtained at each site. Total NRMs were generally well grouped, but in a few cases large randomly directed magnetically soft components were removed in low alternating fields, enabling a well grouped stable remanence to be isolated. The criterion for optimum magnetic cleaning is that of Briden[14]. In all cases, progressive a.f. demagnetisation experiments have shown the observed remanences to be of high coercivity and in most cases directionally stable to peak alternating fields > 100 mT. Microscopic examination of polished specimens generally shows euhedral–subhedral magnetite grains with lamellar intergrowths of ilmenite, a textural combination associated with high temperature (> 600 °C) petrogenesis[11]. In spite of the unavailability of baked contact tests or other geological evidence for the age and origin of remanence, the stable NRM is therefore likely to be a TRM acquired during initial cooling. With the exception of the results from the Myndd Nefyn granodiorite, which may be deviant due to slight relative movement in an area of undeniably complex faulting[9], the *in situ* remanences are well grouped with an overall mean of $D=143°$, $I=-71°$ corresponding to a conventional palaeomagnetic pole at 108°E, 68°S.

This mean pole is nearly 90° away from published Ordovician–Devonian poles for Britain and an explanation must be sought.

Inevitably there are time gaps in the palaeomagnetic record, and it may well be that the new data fall in one of these. Gaps of the order 10^7 yr are to be expected given the sparseness of published data. Therefore if the discrepancy is to be explained in terms of polar shift on the assumption of an axial geocentric dipole field, then apparent polar wander at an average rate approaching 20° Myr^{-1} has to be invoked. This is an order of magnitude greater than apparent polar wander arising from plate motions in more recent geological times. Even if it were accomplished by minimum lateral motion (given by the palaeolatitude change) together with rotation about a vertical axis close to the sampling area (to accomodate the declination discrepancy), motion of > 50 cm yr^{-1} is required, which is still excessive on any plate tectonic analogy.

An interval of 10^7 yr might also be thought reasonable for the emplacement of these widely spaced North Wales bodies. It is hard to imagine an overall intrusion time of < 10^6 yr for such a province. Hence the possibility that these rocks record a single geomagnetic polarity transition is effectively excluded because estimates on Tertiary to Recent transition times indicate that they typically take 10^3–10^4 yr. Nevertheless it is intriguing to note that the pole position from the North Wales studies is almost equatorial to the typical Ordovician palaeomagnetic field. This is in common with the finding of Shaw[12] that the field paused in this half-way state during a polarity transition recorded in the palaeomagnetism of some Icelandic rocks.

In an area of such complex structural history as North Wales it might be expected that the *in situ* remanences of the intrusions require adjustment for tilt and/or local rotation. This does not, however, seem to be the case, as these massive intrusions, many of which have near vertical 'stock-like' shapes, would require an unlikely 60° northward rotation about a horizontal axis to remove the palaeomagnetic discrepancy. Also, they are seen to carry near vertical contemporaneous dikes. Table 1 and Fig. 1 also contain the tilt-adjusted palaeomagnetic results from the structurally simple Ordovician Upper and Lower Basic Lavas of Cader Idris. Their overall mean magnetisation falls within the group of *in situ* results from the intrusions. The magnetic stability of the lavas was comparatively low, but the similarity of the two results does reinforce the suggestion that the intrusions have suffered little tectonism. Therefore unless evidence is produced to show that the bodies sampled in this study, which are geographically

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widely scattered from Cader Idris to Conway, have all been rotated by the same amounts by a bodily movement of North Wales, then their *in situ* remanences must be considered to define the geomagnetic field at the time when those remanences were acquired.

In North Wales there is no geological evidence of any large scale post-Caledonian thermal event at high enough temperatures to explain the stable remanent magnetisations observed in this study. In any case the only known palaeomagnetic field corresponding to these results is that of Tertiary to Recent times. The small amount of Tertiary igneous activity in North Wales could not have been responsible for magnetising rocks on the large scale necessary to explain the observations.

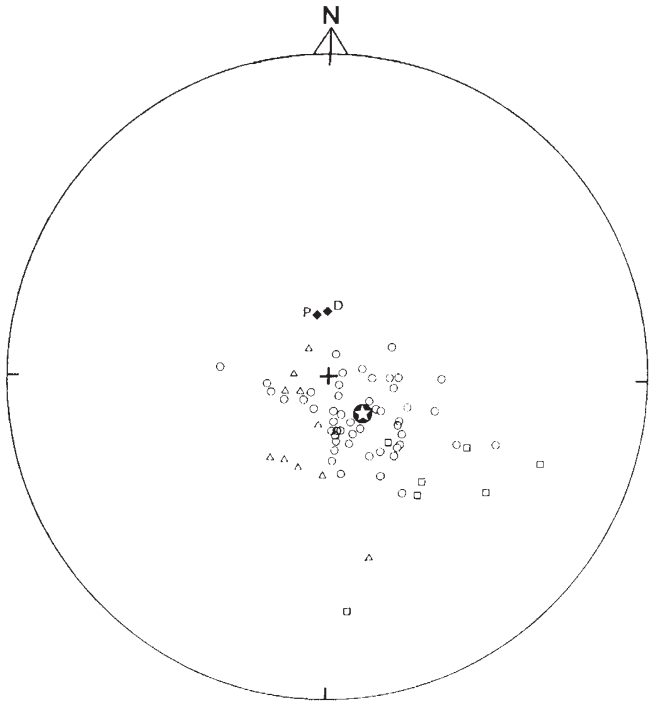


Fig. 1 Stereographic projection of site mean remanences after a.f. cleaning of intrusives from North Wales, and the basic lavas of Cader Idris. ○, *In situ* remanences in all intrusive rocks except Myndd Nefyn; the star indicates their overall mean direction; □, the *in situ* results from Myndd Nefyn; △, tilt-adjusted results from the basic lavas of Cader Idris. The present axial geocentric dipole and the present geomagnetic field directions are marked ◆ and labelled D and P respectively. Open symbols indicate negative (upwards) inclinations and closed symbols positive (downwards) inclinations.

All conventional interpretations of the results of this study are therefore unsatisfactory, and instead, the possibility of an anomalous geomagnetic field lasting several million years during the Lower Palaeozoic must be seriously considered. It is most likely that we have recorded only one such anomalous event in this study because of the improbability of successive intrusions repeatedly occurring at similar geomagnetically anomalous instants. Evidence of precisely when this anomaly might have existed is inconclusive. Its occurrence in late Ordovician times would, however, be compatible with the inferred primary TRM of the intrusions, their possible Ordovician age and the similarity of their remanence directions with that of the proven Ordovician Cader Idris lavas.

The reality of an anomalous magnetic field lasting ~ 10^7 yr would be of considerable geomagnetic interest. Whether it exists or not should be capable of resolution by palaeomagnetic studies elsewhere in the world.

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Palaeomagnetism of a slowly cooled plutonic terrain in western Greenland

Palaeomagnetic results from a high grade Precambrian terrain in western Greenland show many features attributable to magnetisation during very slow cooling caused by gradual uplift and erosion. They may locate a whole segment of the Laurentian apparent polar wander (APW) path with unusually high precision and also indicate the probable direction of movement.

The magnetic effects described here are analogous to those experienced by radiometric systems during slow cooling[1,2], and are attributable in part to the fact that magnetic minerals have a range of blocking temperatures at which the magnetisation becomes effectively fixed. The effective blocking temperatures of magnetic grains will be considerably lowered during slow cooling, possibly to as low as 200 °C (refs 4 and 5) so that magnetisation would be acquired during the later stages of unroofing, when moderate erosion rates can be expected. Average present-day erosion rates have been variously estimated at 30 and 76 m Myr^{-1} (refs 6 and 7, respectively), and if a slightly higher rate of 100 m Myr^{-1} is assumed together with a geothermal gradient of 20 °C km^{-1}, and a blocking temperature range of 50 °C for stably magnetised grains, then the magnetisation of any one specimen would take 25 Myr. Also, because the magnetisation will be acquired as the rocks pass through the (presumably horizontal) isotherms corresponding to their critical range of blocking temperatures, rocks at higher structural levels will become magnetised earlier than those at lower levels. For example, with an erosion rate of 100 m Myr^{-1}, a difference in level of 2 km would correspond to a difference in the mean time of magnetisation of the order of 20 Myr.

Slow cooling during unroofing should therefore produce several effects in the palaeomagnetic record.

- Apart from the effects of APW all of the rocks formed before uplift should be magnetised in the same direction and give the same pole position, even though geological evidence indicates that they are of different ages.
- Because secular variation should effectively be averaged out within each specimen, between-site scatter attributable to this cause should be small.
- If APW occurred during the magnetisation period then poles from sampling sites at different original structural levels should be strung out along a trend corresponding to that particular section of the APW path, with higher sites at the older part of the trend and lower sites at the younger part.

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